**SIMATS ENGINEERING**
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**Course Code**: CSA1616 **Slot:** D

**Course Name**: Data warehouse and Data mining

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**EXP:14 Two Maths teachers are comparing how their Year 9 classes performed in the end of year exams. Their results are as follows: Class A: 76, 35, 47, 64, 95, 66, 89, 36, 8476,35,47,64,95,66,89,36,84**
**Class B: 51, 56, 84, 60, 59, 70, 63, 66,5051,56,84,60,59,70,63,66,50**
**(i) Find which class had scored higher mean, median and range.**
**(ii) Plot above in boxplot and give the inferences Class B: 51, 56, 84, 60, 59, 70, 63, 66, 5051,56,84,60,59,70,63,66,50**

**Aim:**

To compare the performance of Class A and Class B using mean, median, and range, and to visualize the marks using a boxplot in R.

**Procedure:**

1. Enter the examination marks of Class A and Class B into R.
2. Calculate the mean, median, and range of both classes using the appropriate functions.
3. Compare the statistical measures to identify which class performed better.
4. Use the boxplot() function to plot the marks of both classes.
5. Observe the median, spread, and possible outliers in the boxplot and interpret the results.

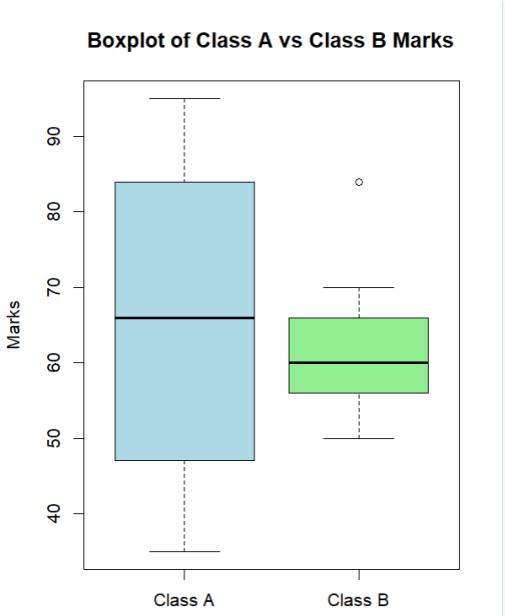
**Code:**

```
1
2  classA <- c(76, 35, 47, 64, 95, 66, 89, 36, 84)
3  classB <- c(51, 56, 84, 60, 59, 70, 63, 66, 50)
4  mean(classA)
5  median(classA)
6  max(classA) - min(classA)
7  mean(classB)
8  median(classB)
9  max(classB) - min(classB)
10 boxplot(classA, classB,
11         names = c("Class A", "Class B"),
12         main = "Boxplot of Class A vs Class B Marks",
13         ylab = "Marks",
14         col = c("lightblue", "lightgreen"))
```

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**Output:**

```
> classA <- c(76, 35, 47, 64, 95, 66, 89, 36, 84)
> classB <- c(51, 56, 84, 60, 59, 70, 63, 66, 50)
> mean(classA)
[1] 65.77778
> median(classA)
[1] 66
> max(classA) - min(classA)
[1] 60
> mean(classB)
[1] 62.11111
> median(classB)
[1] 60
> max(classB) - min(classB)
[1] 34
> boxplot(classA, classB,
+         names = c("Class A", "Class B"),
+         main = "Boxplot of Class A vs Class B Marks",
+         ylab = "Marks",
+         col = c("lightblue", "lightgreen"))
```




**Result:**

The mean, median, and range of both classes were calculated successfully, and the boxplot was generated in R. Class A has a higher mean and median, while Class B has a smaller range, indicating more consistent marks.

**EXP:15 Let us consider one example to make the calculation method clear. Assume that the minimum and maximum values for the feature F are $50,000 and $100,000 correspondingly. It needs to range F from 0 to 1. In accordance with min-max normalization, v = $80, Use the two methods below to normalize the following group of data: 200, 300, 400, 600, 1000**
**(a) min-max normalization by setting min = 0 and max = 1 (b) z-score normalization.**

**Aim:**

To perform min–max normalization and z-score normalization on the given dataset using R

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**Procedure:**

1. Enter the dataset containing the values 200, 300, 400, 600, and 1000.
2. Find the minimum and maximum values using min() and max().
3. Apply min–max normalization to scale the values between 0 and 1.
4. Calculate the mean and standard deviation using mean() and sd().
5. Apply z-score normalization and display the normalized values.

**Code:**

```
1
2  x <- c(200, 300, 400, 600, 1000)
3  min_x <- min(x)
4  max_x <- max(x)
5  minmax <- (x - min_x) / (max_x - min_x)
6  print(minmax)
7  mean_x <- mean(x)
8  sd_x <- sd(x)
9  zscore <- (x - mean_x) / sd_x
10 print(zscore)
```

**Output:**

```
> x <- c(200, 300, 400, 600, 1000)
> min_x <- min(x)
> max_x <- max(x)
> minmax <- (x - min_x) / (max_x - min_x)
> print(minmax)
[1] 0.000 0.125 0.250 0.500 1.000
> mean_x <- mean(x)
> sd_x <- sd(x)
> zscore <- (x - mean_x) / sd_x
> print(zscore)
[1] -0.9486833 -0.6324555 -0.3162278  0.3162278  1.5811388
```

**Result:**

Min–max normalization and z-score normalization were performed successfully in R. Min–max normalization scaled the values between 0 and 1, while z-score normalization transformed the values based on their mean and standard deviation.

**EXP:16 Obtain Multiple Lines in Line Chart using a single Plot Function in R.Use attributes“mpg”and“qsec”of the dataset “mtcars”.**

**Aim:**

To plot multiple lines in a single line chart using the mpg and qsec attributes of the mtcars dataset in R.

**Procedure:**

1. Load the built-in mtcars dataset in R.
2. Use the plot() function to draw a line chart for the mpg attribute.
3. Add the qsec attribute to the same chart using the lines() function.
4. Assign different colors to distinguish the two lines.
5. Add a legend using the legend() function and observe the chart.

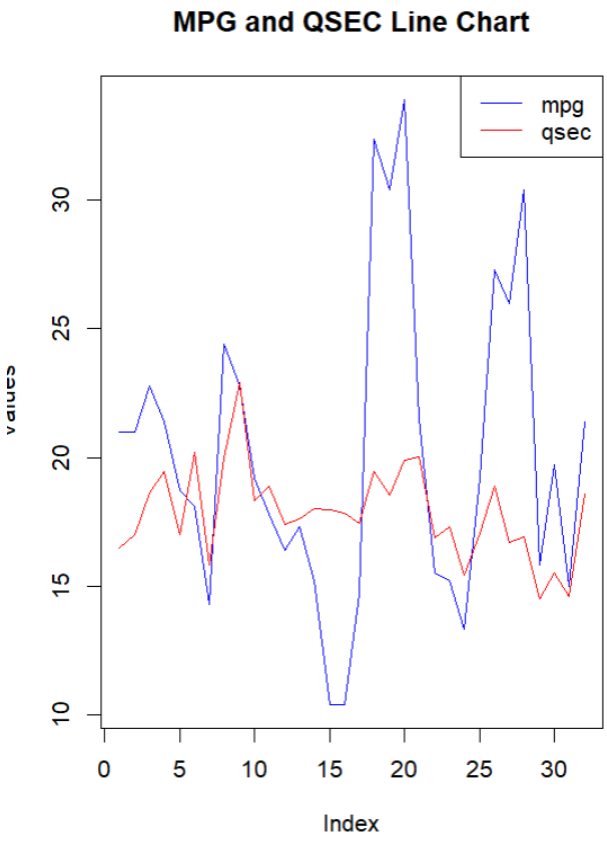
**Code:**

```
plot(mtcars$mpg,
     type = "l",
     col = "blue",
     ylab = "Values",
     xlab = "Index",
     main = "MPG and QSEC Line Chart")


lines(mtcars$qsec, col = "red")


legend("topright",
       legend = c("mpg", "qsec"),
       col = c("blue", "red"),
       lty = 1)
```

**Output:**




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**Result:**
Multiple line graphs representing the mpg and qsec attributes of the mtcars dataset were plotted successfully on a single chart in R using different colors and a legend.

**EXP:16** Make a histogram for the "AirPassengers "dataset, start at 100 on the x-axis, and from values 200 to 700, make the bins 150 wide

**Aim:**
To plot multiple lines in a single line chart using the mpg and qsec attributes of the mtcars dataset in R.

**Procedure:**

1. Load the built-in mtcars dataset in R.
2. Use the plot() function to draw a line chart for the mpg attribute.
3. Add the qsec attribute to the same chart using the lines() function.
4. Assign different colors to distinguish the two lines.
5. Add a legend using the legend() function and observe the chart.

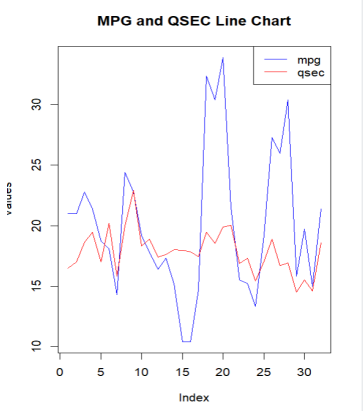
**Code:**

```
plot(mtcars$mpg,
     type = "l",
     col = "blue",
     ylab = "Values",
     xlab = "Index",
     main = "MPG and QSEC Line Chart")


lines(mtcars$qsec, col = "red")


legend("topright",
       legend = c("mpg", "qsec"),
       col = c("blue", "red"),
       lty = 1)
```

**Output:**




**Result:**
Multiple line graphs representing the mpg and qsec attributes of the mtcars dataset were plotted successfully on a single chart in R using different colors and a legend.

**EXP:17 Make a histogram for the "AirPassengers "dataset, start at 100 on the x-axis, and from values 200 to 700, make the bins 150 wide**

**Aim:**

To create a histogram for the AirPassengers dataset in R by setting the x-axis starting point to 100 and using a bin width of 150 up to 700.

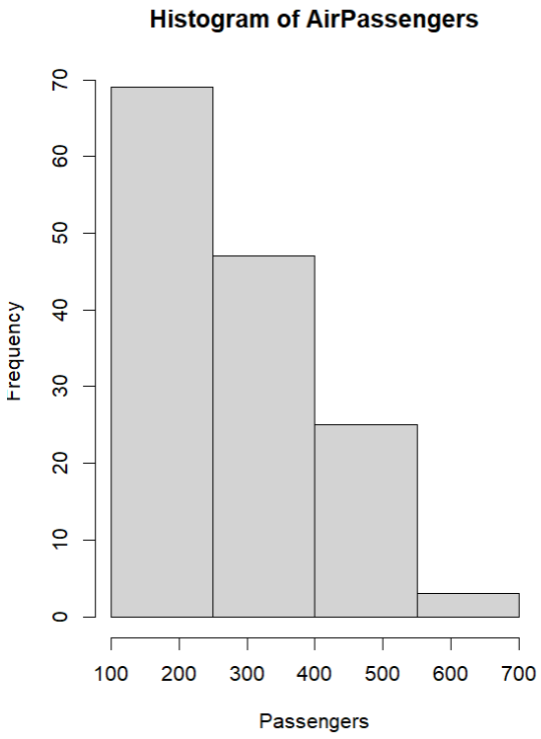
**Procedure:**

1. Load the built-in AirPassengers dataset in R.
2. Store the dataset in a variable.
3. Set the histogram breakpoints from 100 to 700 with an interval of 150.
4. Use the hist() function to generate the histogram.
5. Add a suitable title and x-axis label to the histogram.

**Code:**

```
data <- AirPassengers
breaks <- seq(100, 700, by = 150)
hist(data,
     breaks = breaks,
     main = "Histogram of AirPassengers",
     xlab = "Passengers")
```

**Output:**




**Result:**

Multiple line graphs representing the mpg and qsec attributes of the mtcars dataset were plotted successfully on a single chart in R using different colors and a legend.

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